LAB 10

Task 1 :

```
% Square
square(Side, Area, Perimeter) :-
    Area is Side * Side,
    Perimeter is 4 * Side.

% Rectangle
rectangle(Width, Height, Area, Perimeter) :-
    Area is Width * Height,
    Perimeter is 2 * (Width + Height).

% Triangle
triangle(Base, Height, A, B, C, Area, Perimeter) :-
    Area is 0.5 * Base * Height,
    Perimeter is A + B + C.
```

```
101 ?- consult(['c:/Users/alhan/OneDrive/سطح المكتب/task1.pl']).
true.

102 ?- square(5, Area, Perimeter).
Area = 25,
Perimeter = 20.

103 ?- rectangle(4, 6, Area, Perimeter).
Area = 24,
Perimeter = 20.

104 ?- triangle(3, 4, 3, 4, 5, Area, Perimeter).
Area = 6.0,
Perimeter = 12.
```

Task 2:

```
% User Dataset
user(admin, 1234).
user(student, 5678).
user(alhanouf, 1111).

% Start Login
login :-
    login_attempt(3).

% Check Login
login_attempt(Attempts) :-
    Attempts > 0,
    write('Enter Username: '),
    read(Username),
    write('Enter Password: '),
    read(Password),
    (
        user(Username, Password)
    ->
        write('Login Successful'), nl
    ;
        write('Login Unsuccessful'), nl,
        Remaining is Attempts - 1,
        login_attempt(Remaining)
    ).

% Maximum Attempts Exceeded
login_attempt(0) :-
    write('Sorry, you exceeded the maximum number of attempts, try again later.'),
    nl.
```

```
101 ?- consult(['c:/Users/alhan/OneDrive/سطح المكتب/task2.pl']).
true.

102 ?- login.
Enter Username: admin
|: .
Enter Password: |: 1234.
Login Successful
true.

103 ?- login.
Enter Username: alhanoof
|: .
Enter Password: |: 1234.
Login Unsuccessful
Enter Username: |: alhanoof.
Enter Password: |: 1234.
Login Unsuccessful
Enter Username: |: alhanoof.
Enter Password: |: 1234.
Login Unsuccessful
Sorry, you exceeded the maximum number of attempts, try again later.
true.
```

Task 3:

We can use Prolog operators and rules in our practical project to perform calculations, validate user inputs, and make decisions based on specific conditions.